Unexplored and familiar: experiencing interactive spatio-temporal visualization

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Abstract — **Spatio-temporal (ST) visualization has the power to convey complex information, allowing the user to observe, identify, and interpret occurrences and the evolution of events. A comparative case study was designed to assess the predominantly privileged strategies to represent the articulation between temporal variation and geographic data. The systematic analysis of 25 cases of ST visualization has led to the identification and acknowledgment of shared characteristics between them, and the distinction of disparities and exceptions. The application of 50 criteria, assorted in 11 categories, resulted in nine general findings and two startling remarks. This study provides an understanding of the most prevalent author's decisions when designing interactive ST visualization.**

Keywords - spatio-temporal data; information visualization; interactivity; interface design; case study.

I. Introduction

Spatio-temporal (ST) visualization has the power to convey complex information. It not only allows the user to observe, identify, and interpret events (or their evolution) but also to comprehend them through its localization in both time and space. The potential of ST visualization is relevant because of the diversity of its possible areas of application, such as data journalism, educational tools, traffic flow analysis, observation of climate change, monitorization of migratory flows, among others. When interactive, ST visualization tools promote the exploration, discovery, and analysis of complex messages. Generically, ST visualization allows the user to better understand the overall image of a given reality at a given moment (or time interval), and relate it to its own time and space, at different scales: locally, regionally or globally, and with the past, present, and future.

This research focuses on graphical interfaces that consider the *geographical space* the scenario of the communicative action. Simultaneously, the *temporal variation* of the information reveals changes in one or more objects or themes. Furthermore, the *interactive dimension*, allows the user to engage with the system, by adjusting the object for a customized exploration, through filters, zoom, visualization encoding, among others. Firstly, an extensive literature review on these dimensions allowed a compilation of guidelines and reflections on best practices in visualizing ST information. These premises were ground base for the establishment of the criteria of the present analysis. Then, a comparative case study was conducted, aiming to establish the predominantly privileged strategies to visualize and interact with the articulation between temporal variation and geographical location of data. This method was adopted, assuming that covering multiple cases would raise generalizable knowledge about the collection. The longitudinal and systematic analysis of 25 examples of ST visualization of reference has led to the identification and acknowledgment of shared characteristics between them, and the distinction of disparities and exceptions. The elected research method aims to mitigate the gap of this kind of approach regarded in previous studies on ST visualization.

II. Background

Research conducted by Rodrigues, Figueiras, and Alexandre [1] has led to infer that the design of ST visualization implies an elaborate process, motivating countless decisions of different nature. This research gathered and analyzed which requirements and dimensions should be considered throughout its design process. Thereafter it was assumed that a study on the currently available examples of ST visualization would contribute to answering a subsequent research question: "How can these requirements and dimensions effectively integrate a graphical interface?". The review, synthesis, and systematization of these requirements and dimensions originated eleven operative categories of analysis for the present study: *Data properties; Intended experience; Purposive interpretation; Spatio-temporal phenomena; Interrogative framework; Type of map; Temporal structure; Temporal scale; Spatial scale; Interaction techniques; and Narrative and storytelling.*

A. Data properties

The first moment of incidence of this analysis focuses on the data set applied, and the identification of its properties and implications. A correct understanding of the data set's nature is the foremost step on which all the following decisions depend [2]. According to Martinelli [3], in the context of thematic cartography, three classes of relationships between objects or spatial positions may occur within an ST data set. In the case of a *relationship of diversity* – also called nominal or categorical – it is portrayed the quality of the existence, nature, species, location, or occurrences' extent of a phenomenon (or its attributes). On the other hand, a *relationship of order* outlines hierarchies or relative positions between aspects of the spatial sets, arranged in a logical and clear sequence. Finally, a *relationship of proportionality* is portrayed when the focus is on the comparison of quantitative values between specific properties of the objects. Concurrently, the quality of the

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measurement can also define the data set [4]. *Discrete data* can only take distinct values, which are independent of each other. In opposition, *continuous data* implies a continuity between the individual values. Besides, regarding the dimension of the data, these are rated *absolute* when they expose the total values measured, and *relative* when they seek to normalize the data so that the different objects are comparable to each other, as in the case of rates or ratios.

B. Intended experience

The structures suggested by several authors [5, 6, 7, 8, 9] regarding the communication modes, allowed us to reason that there are two primary intentions of user experience in ST visualization. In the case of a presentation of the information that follows an editorial purpose structured a priori, we consider the intention to be the *communication* of a message. On the other hand, if no guidance is given on the possible conclusions to be drawn from the interaction with the visualization, the intention is open discovery. In this scenario, free of a prior narrative organization of information, it is determined that the intention of the experience is *exploration*.

C. Purposive interpretation

According to Kirk [7], information may be visualized aiming to accomplish three distinct main purposes of interpretation. The interpretation seeks *comparisons and proportions* when it is intended, for example, to learn the range of values presented or the means they are distributed in space and time; or to learn about the order of magnitude between different objects, or understand the values presented in the context of averages, targets, and forecasts. On the other hand, if the user can observe the direction and the data evolution trend, evaluate whether this evolution is linear or exponential, or identify consistent patterns, fluctuations, or intersections, the interpretation purpose is summed as *trends and patterns*. Finally, in the case of the interpretation of *relationships and connections*, it is possible to distinguish exceptions, determine correlations between variables, identify clusters and gaps, or establish the distribution or hierarchical relationships between categories of data.

D. Spatio-temporal phenomena

To visualize temporal variation in space, ST data may be classified according to three types of phenomena: *existential changes, spatial changes,* and *thematic changes* [10, 11]. The first one represents the appearance and disappearance of objects on the map. Next, in order, spatial changes represent variations of movement of objects in space. Lastly, in the context of thematic changes, it is feasible to analyze variations in the values of the attributes, i.e., there occur changes in the thematic properties.

E. Interrogative framework

When analyzing ST data, the user can simultaneously pose three kinds of questions, considering the object (what), location (where), and time (when). According to Peuquet's framework [12], the first interrogative approach focuses on the identification of objects (or thematic properties) existing at a given place and time, and formalizes it as *"When + Where > What"*. The second question highlights the location occupied by a given object (or thematic value) at a given time, translated into the axiom *"When + What > Where"*. Ultimately, a third way of structuring an interrogative stance in ST visualization examines the time interval or the frequency that a given object (or thematic value) occupies a particular place, and can be expressed as *"Where + What > When"*.

F. Type of map

For effective communication, the choice of the thematic map that best accommodates the purpose of the visualization should also be a conscious decision. Throughout this study, nine different types of thematic maps were regarded: *dot distribution map, proportional symbol map, choropleth map, contour maps, cartogram, network map, flow map, Voronoi map, and 3D map.*

G. Temporal structure

Aigner, Miksch, Schumann, and Tominski [13] identify three types of temporal structures in information visualization: ordered time, branched time, and multiple perspectives. For this study, we only considered the ordered structure because it appears to be the most commonly used [14]. Within the ordered time, it is common to verify a continuous sequence of occurrences, from past to future, exhibiting *linear time*. On the other hand, if the events occur periodically, the structure reveals the *cyclic time*.

H. Temporal scale

The temporal dimension is composed of two primitives: time points and time intervals. A point represents one instant of time, while a time interval holds a certain extent [15]. These primitives can be translated into different time-scale granularities, such as seconds, minutes, hours, months, years, centuries, among others. In interactive information visualization, the temporal scale can be *fixed*, maintaining the granularity of the elected time unit constant, or also the speed of the animation. On the other hand, the system may provide a *manipulable* analysis by allowing the variation of the granularity of the scale units, regulating the pace at which the information is displayed, enabling to pause or even to move the cursor along the timeline. These adjustments are crucial for the identification of the appropriate circumstances that best suit an effective understanding of the information [14].

I. Spatial scale

The choice of spatial scale plays an important role in visualization since it affects the quantity and size of graphic elements represented, i.e., the level of detail displayed on the map [15, 16]. Similarly to what occurs at defining the temporal scale, the geographic space can be sized in different granularities: at the scale of one street, neighborhood, district, country, continent, or globally. The determination of the point of view is seldom *fixed*, not allowing variations on the map. Nevertheless, the experience may be enhanced through the inclusion of interaction tasks to zoom in and out one area of interest, allowing the user to *manipulate* the degree of nearness and settle it for the analysis purpose.

J. Interaction techniques

The process of gaining insights through an exploratory discovery in the visual communication of information is enhanced by the convenience of interactivity [17]. The incorporation of interaction tasks grants several advantages, namely by increasing the overall control of the experience

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through the potential of its customization [18]. For the study here presented, we adopted the typology suggested by Figueiras [19] as criteria for interactive techniques. The selected typology results in a comprehensive compilation of 11 interaction techniques*: filtering; selecting; abstract/elaborate; overview and explore; connect/relate; reconfigure; encode; history; extraction of features; participation/collaboration; and gamification.* These consolidate the main contributions of the last decades on this topic, namely Shneiderman's Mantra [20], distinctive taxonomies [21, 22], and the levels of interaction by Mazza [6].

K. Narrative and storytelling

Generally, stories are constructions based on the fundamentals of change and movement. An effective way to create a story from data is to consolidate a verbal narrative that articulates the dimension of time with the respective spatial dimension. As in other genres of data stories, the incorporation of narrative elements in ST visualization plays a crucial role in enhancing the communication process, helping users to explore and interpret the intended message [7, 17].

The last category of analysis gathers the resources that ensure the understanding of the visualization and assist in the exploratory interaction. Firstly, when using a map, it is essential to incorporate *auxiliary cartographic elements*, such as title, subtitles, legend, orientation, accessory maps, captions, data source, publication date, credits, and technical information [3]. In turn, *additional narrative elements*, such as introduction, visual annotations, and small animations, may improve the storytelling experience [7]. It may also be advantageous to complement the visualization with *auxiliary graphs* that provide other perspectives on the topic, such as histograms, line charts, tables, map navigator, among others. Furthermore, *usability features*, such as user instructions, error prevention, alerts, inter alia are acknowledged for raising a more satisfying experience [23]. A precise, responsive, and efficient system enhances the perception of competence, mastery, and control of the user [24]. Lastly, we have also considered the analysis criterion *sound* as a supportive experience resource, either in the form of narration or in the attribution of sounds to specific occurrences.

III. METHODOLOGY

In order to recognize what attributes graphical interfaces in ST visualization have in common, and what disparities and exceptions can be detected, we conducted an exhaustive analysis of a set of 25 reference cases. The preference for this method relies on the potential of an empirical approach regarding existing subjects, held in their natural context, applying multiple forms of evidence [25]. This method was chosen because case study research is generally more exploratory than confirmatory, i.e., the researcher seeks to identify themes or categories of behaviors and events, instead of testing hypotheses. The comparative case study aims to analyze a given sample of several cases, parts, or members of one whole. When considering multiple cases, it is assumed that each case has its characteristics and queries. The objective is to understand how the whole behaves, and in what way specific situations are solved throughout the different cases [26].

In this research, we have defined the central topic of interest in the primary stage of the study: the literature review assisted the process by establishing a relevant and viable background theoretical framework. This way, it was possible to determine what was known at the beginning, i.e., that ST visualization is (1) complex, since the development of this type of systems implies numerous design decisions of different levels of complexity which are dependent on each other; (2) multidisciplinary, because its design inevitably requires the intersection of different areas of expertise; and (3) scarce, considering a gap was detected between its broad communicative potential and the effective frequency of usage of this kind of visualization.

An operative table was designed: each column accommodates one case, and the rows cover the 50 criteria of analysis, assorted in 11 categories as described in the Background section. The sequence in which we analyzed the criteria followed the natural succession of the decision-making process, although the design process is intrinsically not linear.

The application of the categories attempts to identify specific characteristics of each visualization and understand how each criterion varies throughout the 25 cases, as shown in Figure 1. Every case was analyzed individually on a MacBook Air connected to an external 28" monitor Samsung LU28E570DS (2560x1440px), using the Google Chrome V. 79.0 .3945.117 browser, and with sound output on.

A. Cases of study

The analyzed 25 cases of interactive ST visualization were selected according to one prevailing convenience criterion: their availability on the web. Also, the interactive dimension was assumed as a criterion of exclusion, meaning that both static visualizations and animated videos that only provided basic interaction tasks, such as play, pause, and stop, were not contemplated. Thus, we considered that the elementary interaction with an animated video is not representative of the potential that interactivity enhances in the visualization of ST information, and, for this reason, they do not match the defined object of study. The 25 selected cases are:

a. NEO – NASA Earth Observations – 2005 (link)

b. Flightradar24 – 2006 (link)

c. MarineTraffic – 2007 (link)

d. Snow plows in New York City – 2013 (link)

e. NASA Worldview – 2013 (link)

f. Foursquare Time Machine – 2013 (link)

g. HubCab – 2014 (link)

h. The Refugee Project – 2014 (link)

i. In flight – 2014 (link)

j. Child Lives, 2014 (link)

k. U.S. Daily Temperature Anomalies – 2014 (link)

l. World City Populations 1950-2035 – 2015 (link)

m. When Do Americans Leave For Work? – 2015 (link)

n. imagineRio – 2015 (link)

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	a	b	c	d	e	f	g	h	i	j	k	l	m	n	o	p	q	r	s	t	u	v	w	x	y
Data properties																									
relationship of diversity	x	x	x	x	x	x	x	x	·	·	x	·	·	x	x	x	·	x	x	x	x	x	·	·	·
relationship of order	x	x	x	x	x	x	x	x	x	x	x	x	x	·	x	x	x	x	x	x	·	x	·	x	x
relationship of proportionality	x	x	x	·	x	x	x	x	x	x	x	x	x	·	x	x	x	x	x	x	·	x	x	x	x
discrete data	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
continuous data	x	x	x	·	x	·	·	·	·	·	·	·	·	·	x	·	·	·	·	·	·	·	·	·	·
absolute data	x	x	x	x	x	x	x	x	x	x	·	x	·	x	x	x	x	x	x	x	x	x	x	x	·
relative data	x	x	x	·	·	x	·	·	x	x	x	x	x	·	·	·	x	·	x	x	·	x	·	x	x
Intended experience																									
communication	·	·	·	·	x	·	·	x	x	x	x	x	x	·	x	·	x	·	x	x	·	·	x	·	x
exploration	x	x	x	x	·	x	x	·	·	·	·	·	·	x	·	x	·	x	·	·	x	x	·	x	·
Purposive interpretation																									
comparisons and proportions	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
trends and patterns	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
relationships and connections	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Spatio-temporal phenomena																									
existential changes	x	x	x	·	x	x	x	x	x	·	x	·	·	x	x	x	x	x	x	·	x	·	x	x	·
spatial changes	x	x	x	·	x	·	·	·	x	·	·	·	·	·	x	·	·	·	·	·	·	x	·	·	x
thematic changes	x	x	x	x	x	·	·	x	x	x	·	x	x	·	x	·	x	x	x	x	·	x	·	·	x
Interrogative framework																									
When + Where > What	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
When + What > Where	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Where + What > When	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Type of map																									
dot distribution map	x	x	x	·	x	x	x	·	x	·	x	·	·	x	x	·	·	·	·	·	x	x	x	·	x
proportional symbol map	·	·	·	·	·	·	·	x	·	x	·	x	·	·	·	x	x	x	·	·	·	·	·	x	·
choropleth map	·	·	·	·	·	·	·	x	·	·	·	·	x	·	·	·	·	·	x	x	·	·	·	·	x
contour map	x	·	·	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
cartogram	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
network map	·	·	x	·	·	·	·	x	x	·	·	·	·	·	x	·	·	·	·	·	·	·	·	·	x
flow map	·	·	x	·	·	·	x	·	x	·	·	·	·	·	x	·	·	·	x	·	·	x	·	·	x
Voronoi map	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
3D map	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
other	·	·	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
Temporal structure																									
linear time	x	x	x	x	x	x	·	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
cyclic time	·	·	·	·	·	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
Temporal scale																									
fixed	·	·	·	·	·	·	·	·	·	x	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·
manipulable	x	x	x	x	x	x	x	x	x	·	x	·	x	x	x	x	x	x	x	x	x	x	x	x	x
Spatial scale																									
fixed	·	·	·	x	·	x	·	·	·	·	·	·	·	·	·	x	·	·	x	·	·	x	·	·	x
manipulable	x	x	x	·	x	·	x	x	x	x	x	x	x	x	x	·	x	x	·	x	x	·	x	x	·
Interaction techniques																									
filtering	x	x	x	·	x	·	x	x	·	·	·	x	·	x	x	·	·	x	·	·	·	x	x	·	x
selecting	x	x	x	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	x	·	·	·
abstract / elaborate	x	x	x	·	x	x	x	x	x	x	x	x	x	x	x	·	x	x	·	x	x	x	x	x	·
overview and explore	x	x	x	·	x	x	x	x	x	x	x	x	x	x	x	x	x	x	·	x	x	x	x	·	·
connect / relate	x	x	·	·	x	·	·	·	·	·	·	·	·	x	·	·	·	·	·	·	·	·	·	·	·
reconfigure	·	·	x	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
encode	x	x	x	·	x	·	·	·	x	·	·	·	·	·	x	·	x	x	·	·	·	x	·	·	·
history	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·
extraction of features	x	x	x	·	x	x	·	x	·	x	x	x	x	·	·	x	x	x	·	x	x	·	·	x	·
participation / collaboration	·	x	x	·	·	x	·	·	·	·	·	·	·	·	·	x	·	·	·	·	·	x	·	·	·
gamification	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	·	x	·	·	·	·
Narrative and storytelling																									
auxiliary cartographic elements	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
additional narrative elements	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	·	x	x	x
auxiliary graphs	x	x	x	·	·	x	·	x	x	x	x	x	x	·	·	·	x	·	x	x	·	x	x	x	x
usability features	x	x	x	·	x	x	x	·	x	·	x	·	x	x	x	x	x	x	x	·	x	·	·	x	·
sound	·	·	·	·	·	x	·	·	x	·	·	·	·	·	x	·	·	x	·	·	x	·	·	·	·

Figure 1. Study overview.

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o. Shipmap – 2016 (link)

p. The History of Urbanization – 2016 (link)

q. Boom to bust – 2016 (link)

r. E3 – Eruptions, Earthquakes, & Emissions – 2016 (link)

s. Two Centuries of U.S. Immigration – 2016 (link)

t. Es war nicht immer der Osten – 2017 (link)

u. Airbnb is growing fast – 2017 (link)

v. Peak Spotting – 2017 (link)

w. Digital Atlas of Innovation – 2017 (link)

x. London AirBnB bookings, Summer'18 – 2019 (link)

y. Singapore Calling – 2019 (link)

B. *Key premises*

The method consisted of a systematic review of the 25 cases according to the predetermined criteria. As it is expected in one exploratory research approach, it was necessary to rethink and redefine specific inquiry matters of each criterion throughout the process. These iterative adjustments establish the feasibility of the elected research procedures. It is usual for the researcher to adopt several guidelines while synthesizing and interpreting the information collected in a case study. Each guideline allows the continuous improvement of the basic research questions of the study, in light of the data obtained at the beginning of the investigation [27]. This flexibility promotes the understanding of the options regarding the design components and the identification of how practical applications usually occur. For the viability of the analysis, three main premises were established:

- For the verification of the analysis criteria, the entire communicative object was considered, including supplementary text or auxiliary graphs. This premise determines that situations may emerge in which the criteria are not found in the primary map but rather included on complementary tables or charts, e.g., quantitative values to concede the relationship of proportionality;
- It was assumed that the opportunity for users to add comments to the visualization constitutes a technique of participation. On the contrary, sharing on social networks or via email does not: while the first action adds something to the communicative object, the second does not interfere with it;
- We realize that there is an implied level of filtering in the context of temporal visualization since, in most cases, it is possible to choose a singular moment or interval. Nevertheless, in this research, the interaction task *filtering* in ST visualization presupposes tangible actions, such as omitting a category of objects or criteria of representation.

IV. Discussion

The outcome of the case study can be synthesized in a grid of results revealing the answer to the key question "Which criteria were found in which cases?". The overall picture, presented in Figure 1, concedes several interpretations, depending on the focus of interest. For the present research, we have outlined nine general findings and two startling remarks.

A. *General findings*

- Although many of the 25 analyzed visualizations allow zooming, only three (*b.*, *c.* and *e.*) display the map scale, providing the constant information on the ratio of the distances on the map to the corresponding actual distance. In all three cases, the map scale fully adapts to the responsiveness of zooming and the necessary projection adjustments resulting from latitude variation;
- Only five of the analyzed cases include sound. We consider that more visualizations could benefit from the association of sound as an optional narrative element, particularly because these objects are mainly designed for the desktop experience;
- Out of the 25 analyzed cases, 17 opt to complement the experience with auxiliary graphs, improving the message to be conveyed;
- Almost every visualization uses manipulable time and space scales, benefiting of interactivity to enhance the user's engagement;
- A total of 14 cases allows the analysis of three types of relationships of data: *diversity*, *order,* and *proportionality*, confirming the broad communicative potential of interactive ST visualization;
- Out of 25 cases, 16 enable the extraction of features such as the data set, images or short animations;
- Only five of the 25 analyzed cases admit some kind of participation or collaboration to the visualization;
- Only one of the 25 cases analyzed (case *u.*) includes elements of gamification, e.g., a Quiz.
- As anticipated, there is absolute suitability of the conceptual interrogative framework. This reinforces that in every ST visualization, there's the possibility of simultaneously: (1) focusing on the *object* (What) or thematic value, framed in a specific location and time; (2) highlighting the *location* (Where) occupied by a given object or thematic value, at a given moment; and (3) emphasizing the *time* (When) that a given object (or thematic value) occupies in a given location.

B. *Absolute applicability of three purposive interpretations*

We have inferred that the visualization of the temporal and spatial variables necessarily incorporates all three categories suggested by Kirk [7]: *comparisons and proportions; trends and patterns;* and *relationships and connections*. The first one is invariably implicit since, in ST visualization, there is certainly the opportunity to compare between two (or more) different moments, or between the values of two (or more) locations. Secondly, the finding of patterns, fluctuations or intersections is assured by the spatial distribution of information on the map. Additionally, the temporal variation surely highlights fluctuations or trends of the occurrences. Moreover, there is an inevitable association between one temporal moment and the

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corresponding information displayed. In this sense, there is always an exercise of identifying *relationships and connections* between the object and its time, or also through the observation of clusters and gaps of occurrences on the map.

C. Underestimation of the cyclic temporal structure

We observe that only one of the visualizations adopted a cyclic time order *(g. Hubcab)*, in which the volume of taxi rails in NYC is arranged in 8 segments of the 24 hours of one day. Nevertheless, many other analyzed cases would also have profited from a cyclic visualization. In some cases, the periodic data could operate as a complement of the existing timeline as in the cases *a., b., c., e., f., k., o., u.,* and *v.*, revealing yearly fluctuations and trends as supplementary information. On the other hand, in the cases *i., m.* and *y.,* we consider that the cyclic order would constitute a suitable alternative to the employed linear time considering that these data constitute the typical representation of one exemplary interval.

Naturally, this is not always the case. When the main purpose is to observe the evolution of data in a given time interval there is no particular benefit of using the cyclic structure. It is more familiar to apply a linear timeline in visualizations that focus on historical or sequential evolution of events, such as *cases h., n., p., q., r., s.,* and *w.* Similarly, when the time interval represents short periods (as in case *d.* and *x.*) or in the cases where the purpose is to compare between few moments *(j., l.,* and *t.*), it does not make much sense to consider the cyclic structure because it does not match any periodic patterns or fluctuations.

V. Conclusions

Designing ST visualizations arises several challenges from diverse expertise areas. The complexity inherent to ST visualization arises the chance for comprehensive and satisfying experiences. We assumed that its process and implications would be better understood by studying how it is nowadays practiced. The analysis of 50 criteria throughout 25 cases allowed us to reason that interactivity is a recognized resource to enhance the user's engagement with the communicative object. In the context of ST information, interaction tasks proved to support the possibility of identifying different kinds of relationships within the data; allows the manipulation of temporal and spatial scales; and promotes the integration of auxiliary graphs and other narrative elements. Moreover, interaction also supports the increasing concern for the extraction of components.

Finally, we conclude that many visualizations would benefit from the inclusion of extra components. We also observed that the cyclic temporal structure is underestimated and that the inclusion of sound, map scales and tasks for participation, collaboration and gamification could also be increased.

The outcomes of this study intend to constitute the bottom line for the creation of future sets of best practices, or the design of instruments for the assessment of the maturity level of any ST visualization.

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